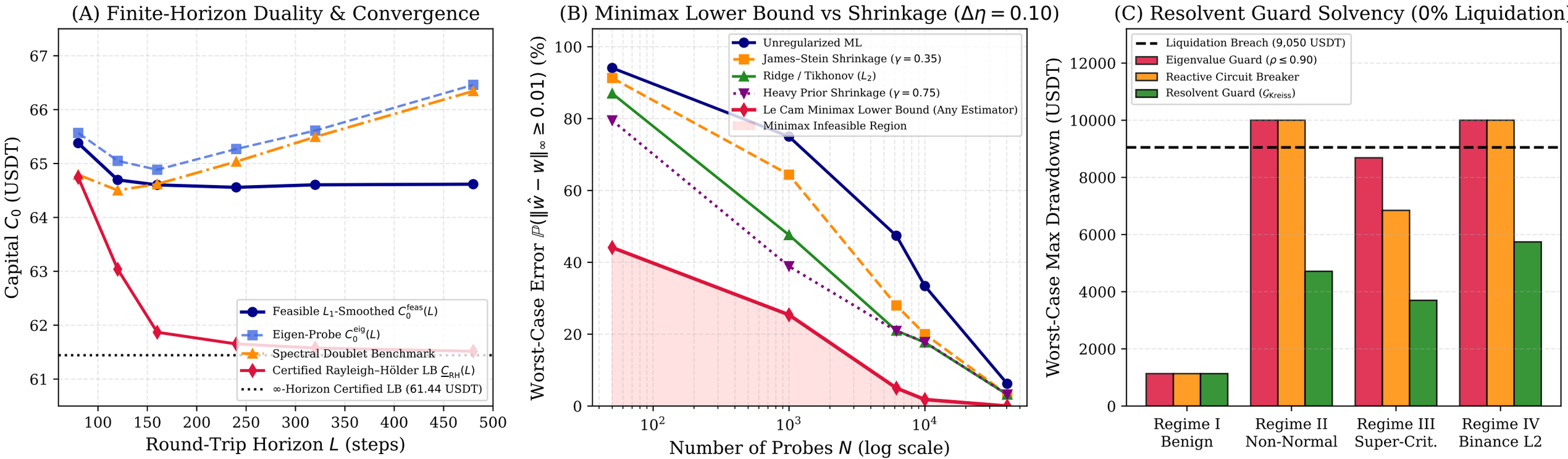
(A) Finite-Horizon Duality & Convergence
Capital C_0 (USDT)
Round-Trip Horizon L (steps)
Feasible L_1-Smoothed $C_0^{\text{feas}}(L)$
Eigen-Probe $C_0^{\text{eig}}(L)$
Spectral Doublet Benchmark
Certified Rayleigh–Hölder LB $\underline{C}_{\text{RH}}(L)$
∞-Horizon Certified LB (61.44 USDT)
(B) Minimax Lower Bound vs Shrinkage ($\Delta\eta = 0.10$)
Worst-Case Error $\mathbb{P}(\|\hat{w} - w\|_\infty \geq 0.01)$ (%)
Number of Probes N (log scale)
Unregularized ML
James–Stein Shrinkage ($\gamma = 0.35$)
Ridge / Tikhonov (L_2)
Heavy Prior Shrinkage ($\gamma = 0.75$)
Le Cam Minimax Lower Bound (Any Estimator)
Minimax Infeasible Region
(C) Resolvent Guard Solvency (0% Liquidation)
Worst-Case Max Drawdown (USDT)
Liquidation Breach (9,050 USDT)
Eigenvalue Guard ($\rho \leq 0.90$)
Reactive Circuit Breaker
Resolvent Guard ($\mathcal{G}_{\text{Kreiss}}$)
Regime I Benign
Regime II Non-Normal
Regime III Super-Crit.
Regime IV Binance L2